PROJECT TECHNICAL REPORT

Customer Segmentation and Churn Prediction Using Machine Learning

---

Prepared By

Student 1	Student 2	Student 3
Name: Salma Abdelhamid	**Name:** Malak Salah	**Name:** Rawda Attia
ID: 230101136	**ID:** 230102892	**ID:** 230102855

Course Information

Course Name: Big Data Analytics
Instructor: Dr. Esraa
Teaching Assistant: Eng. Hend Ahmed
Department: Computer Science Technology

Table of Contents

---

1. Problem Statement

Customer retention is a critical factor in many business applications. Understanding customer behavior and identifying potential churn allows organizations to take proactive actions and improve long-term engagement.

The objective of this project is to build a complete data pipeline that includes:

- Data exploration and preprocessing
- Customer segmentation using clustering techniques
- Churn prediction using machine learning models
- Interpretation of model behavior and performance

The project simulates a real-world customer analytics scenario using structured data resembling a CRM system.

---

2. Dataset Description

The dataset used in this project consists of multiple schemas representing different entities typically found in business environments. The data is synthetically generated using the Python Faker library.

Data link: https://github.com/datablist/sample-csv-files

The dataset is divided into five main schemas:

2.1 Customers Schema

This schema contains customer-level information and serves as the primary dataset for analysis.

Main attributes include:

- Customer ID
- First and Last Name
- City and Country
- Subscription Date
- Contact Information

This dataset is used for both clustering and churn prediction.

2.2 People Schema

This schema includes general user-related information such as:

- User ID
- Name
- Job Title
- Email
- Phone

Although structurally clean, this dataset does not share a common key with the Customers dataset and is therefore not used in modeling.

2.3 Organizations Schema

This schema describes company-level data, including:

- Organization ID
- Name
- Industry
- Country
- Number of Employees

It provides contextual information but cannot be linked to customer records.

2.4 Leads Schema

This schema represents sales-related data:

- Lead Owner
- Company
- Deal Stage
- Source

It reflects business processes but lacks direct integration with customer data.

2.5 Products Schema

This schema contains product-related information:

- Product Name
- Category
- Price
- Availability

Important Note

Although all schemas are stored within the same dataset, they do not share common identifiers. As a result:

- Merging was not feasible
- Each schema was analyzed separately
- Machine learning was applied only to the Customers dataset

---

3. Data Understanding & EDA

Exploratory Data Analysis (EDA) was performed to understand the structure and quality of the data.

3.1 Customers Dataset

Initial inspection showed:

- No missing values
- High cardinality in categorical variables such as City and Country
- Subscription Date stored as text and required conversion

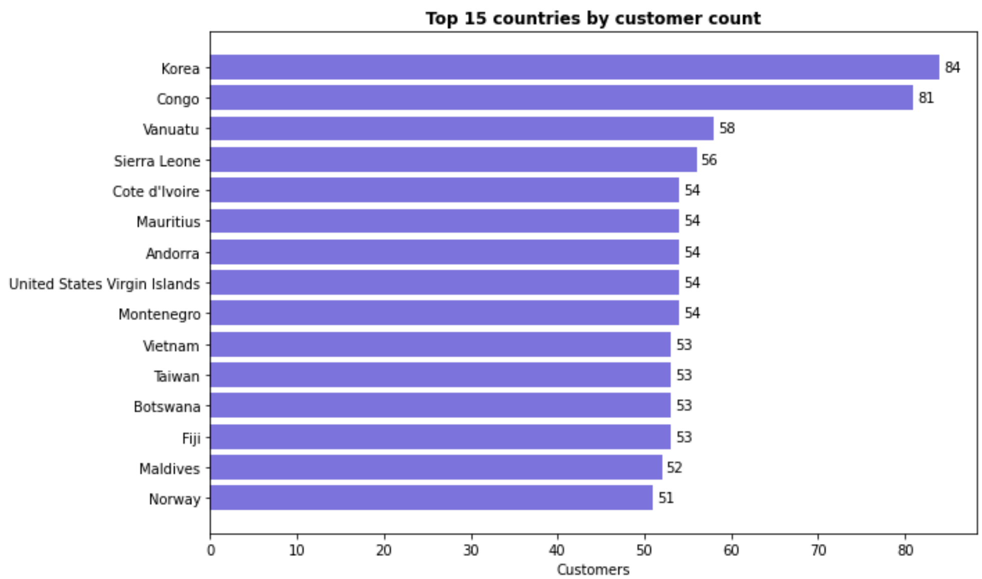

Figure 1: Distribution of Customers by Country

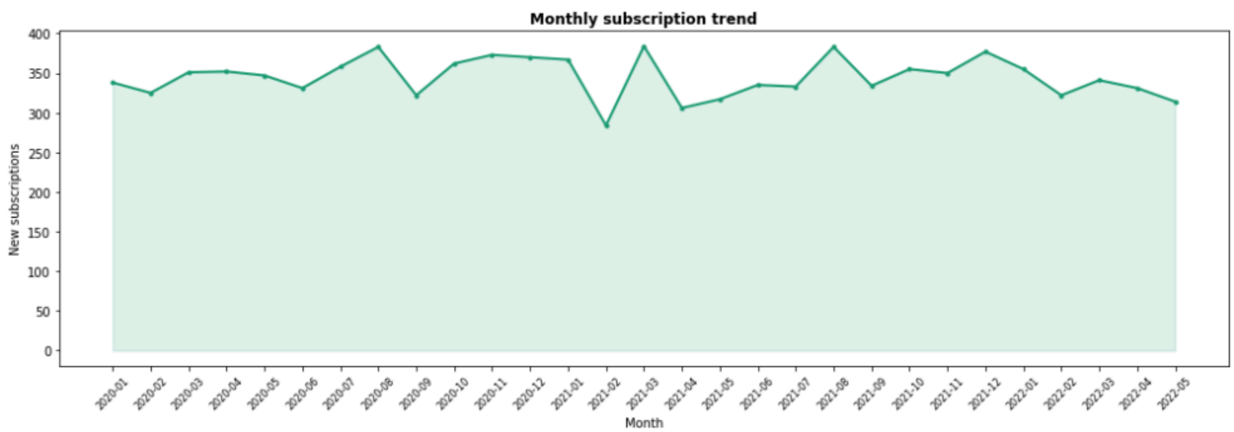

Figure 2: Subscription Trends Over Time

3.2 Other Schemas

EDA revealed that:

- Data is randomly generated
- Most columns contain highly unique values
- No relationships exist between schemas

Conclusion from EDA

The datasets are structurally valid but lack relational consistency. Therefore:

- Only the Customers dataset was selected for modeling
- Other schemas were excluded from the ML pipeline

---

4. Data Preprocessing

Data preprocessing focused on preparing the Customers dataset for analysis.

Steps performed:

- Removal of non-informative columns (IDs, redundant fields)
- Standardization of text fields
- Conversion of Subscription Date to datetime format
- Verification of duplicates and missing values

The preprocessing stage ensured that the dataset is clean, consistent, and suitable for feature engineering.

---

5. Feature Engineering

New features were created to enhance the dataset and improve model performance.

Generated Features

- **days_since_subscription**
  Represents how long a customer has been subscribed
- **recency_group**
  Groups customers into segments based on subscription duration
- **quarter**
  Extracted from subscription month to capture seasonality
- **country_encoded**
  Numerical representation of country values

These features capture both behavioral and temporal aspects of customer activity.

---

6. Customer Segmentation (Clustering)

Clustering was applied to group customers based on behavior and geography.

Method Used

K-Means Clustering

Features Used

- country_encoded
- days_since_subscription

Process

- Data was standardized using StandardScaler
- The Elbow Method was used to determine the optimal number of clusters

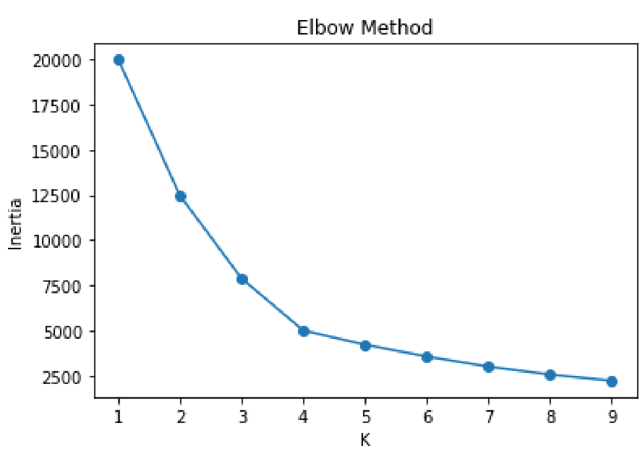

Figure 3: Elbow Method Plot

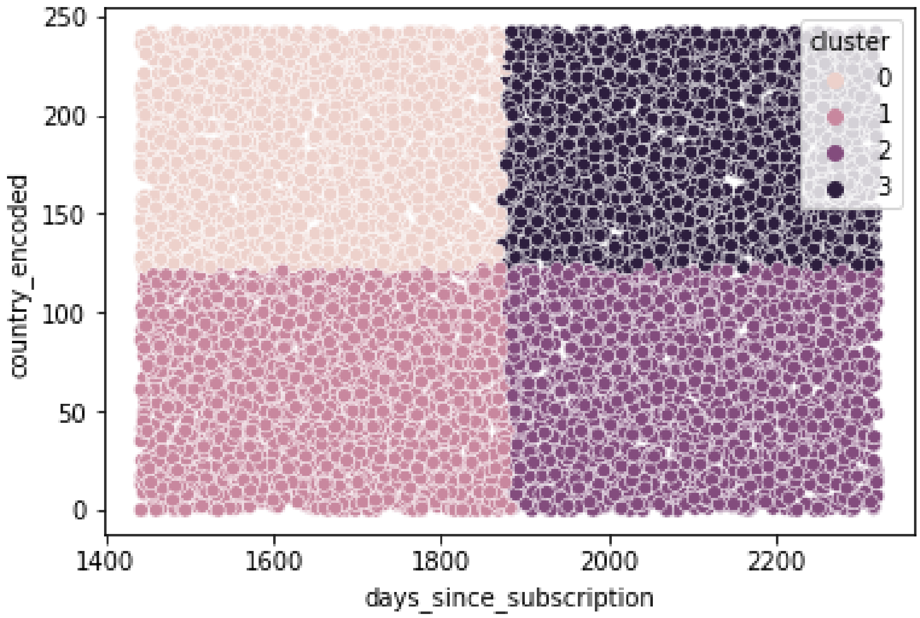

Figure 4: Customer Clusters Visualization

Result

Customers were grouped into four clusters, reflecting differences in:

- Geographic distribution
- Activity level

---

7. Churn Prediction Model

7.1 Problem

The dataset does not contain a churn label.

7.2 Solution

A synthetic churn variable was created based on customer recency, with added randomness to avoid deterministic behavior.

Further explanation for churn (Label Definition and Design Choice) :

Since the dataset does not include a predefined churn variable, a synthetic churn label was created to enable supervised learning. Multiple approaches were explored before selecting the final method.

- ### Approach 1 — Threshold-Based with Randomness

  In the first approach, churn was defined using a fixed threshold on customer activity:

```
customers_fe['churn'] = (
    (customers_fe['days_since_subscription'] > 365) &
    (np.random.rand(len(customers_fe)) > 0.3)
).astype(int)
```

Explanation

- Customers with subscription duration greater than 365 days were considered potential churners
- A random factor was introduced to avoid a fully deterministic rule
- Customers below the threshold were always labeled as non-churn

---

Limitations

This approach introduces a **step-based behavior**, where:

- Customers below 365 days have zero probability of churn
- Customers above 365 days share similar churn probability regardless of how old they are

As a result:

- Important information about customer age is lost
- The model learns a simplified rule instead of a gradual pattern
- The relationship between features and churn becomes less realistic

---

- ### Approach 2 — Continuous Probabilistic Method (Selected)

In the final approach, churn probability was defined as a function of customer recency:

```
prob = customers_fe['days_since_subscription'] /
customers_fe['days_since_subscription'].max()


customers_fe['churn'] = (np.random.rand(len(prob)) < prob).astype(int)
```

Explanation

- Customer age is normalized into a probability between 0 and 1
- Older customers have higher likelihood of churn
- A random process determines the final label based on this probability

---

Advantages

This method provides a more realistic representation of customer behavior:

- Churn probability increases **gradually**, not abruptly
- The full range of the recency feature is preserved
- The model can learn more nuanced relationships

Comparison of Both Approaches

Aspect	Threshold-Based	Probabilistic
Behavior	Step-based	Continuous
Realism	Moderate	High
Information usage	Limited	Full feature utilization
Model learning	Simplified	Richer patterns

Final Decision

The continuous probabilistic approach was selected because it better simulates real-world behavior, where churn likelihood increases over time but is not guaranteed.

This method avoids artificial thresholds and enables the model to learn meaningful patterns from the data.

Defining churn probabilistically improved the quality of the machine learning pipeline by:

- Preventing oversimplified patterns
- Enhancing feature importance interpretation
- Producing more realistic model performance

7.3 Models Used

- Random Forest Classifier
- Logistic Regression

Features Used

- country_encoded
- recency_group
- quarter

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

---

8. Results & Insights

8.1 Random Forest

- Accuracy around 0.67
- Strong performance in detecting churn
- Handles non-linear patterns effectively

Out[61]:

```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', random_state=42)
```

📊 Step 5 — Evaluate

In [62]:

```
from sklearn.metrics import classification_report

y_pred = model.predict(X_test)

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.23      0.34      0.28       370
           1       0.83      0.75      0.79      1630

    accuracy                           0.67      2000
   macro avg       0.53      0.54      0.53      2000
weighted avg       0.72      0.67      0.69      2000
```

8.2 Logistic Regression

- Lower overall accuracy
- Better balance between classes
- More conservative predictions

```
from sklearn.linear_model import LogisticRegression

log_model = LogisticRegression(class_weight='balanced', max_iter=1000)
log_model.fit(X_train, y_train)

y_pred_log = log_model.predict(X_test)

print(classification_report(y_test, y_pred_log))
```

```
              precision    recall  f1-score   support

           0       0.27      0.73      0.39       370
           1       0.90      0.55      0.68      1630

    accuracy                           0.58      2000
   macro avg       0.59      0.64      0.54      2000
weighted avg       0.78      0.58      0.63      2000
```

8.3 Key Observation

Random Forest is more effective for this task, while Logistic Regression provides useful comparison as a baseline model.

8.4 Feature Importance

The most influential features were:

- country_encoded
- recency_group
- quarter (less impact)

Important Insight

Although country appears highly important, this is not necessarily meaningful. Since the dataset is synthetic, this effect is likely due to random distribution rather than true causation.

8.5 Confusion Matrix

- High true positives for churn
- Moderate false positives
- Expected behavior for imbalanced classification

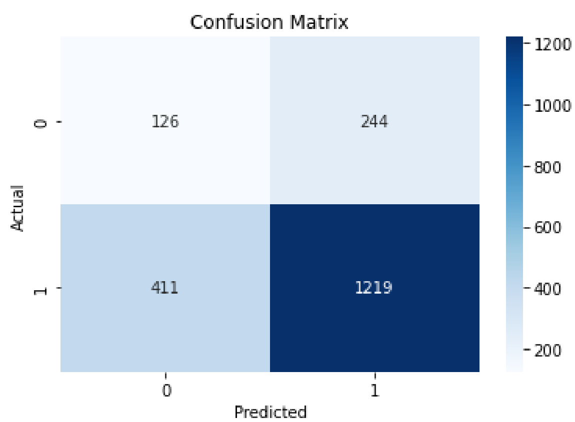

Figure 5: Confusion Matrix

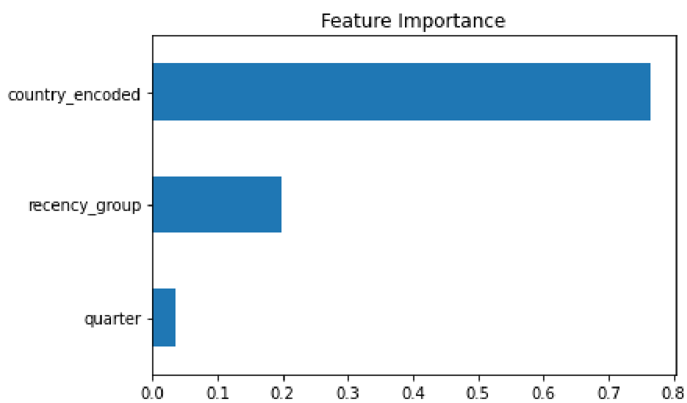

Figure 6: Feature Importance Plot

9. Limitations of the Dataset

The dataset was generated using Python Faker, which introduces several limitations:

- No real-world relationships between variables
- Random distributions across features
- Synthetic churn label
- Potentially misleading feature importance

As a result, the models demonstrate technical performance rather than real business applicability.

10. Conclusion

This project successfully implemented a complete machine learning pipeline, including:

- Data preprocessing
- Feature engineering
- Clustering
- Classification
- Model evaluation

The results demonstrate the ability to extract patterns and build predictive models, even in a synthetic environment.

However, applying this pipeline to real-world data would be necessary to obtain meaningful business insights.
